

Fig 1: One collective, changing interface

The collective is identified with the persistent interior, not a frozen set of boundary variables. Interior nodes (1,2,3) (blue, ringed) are fixed throughout; the screening boundary (orange) is re-inferred and changes identity as z sweeps $-1 \rightarrow +1$.

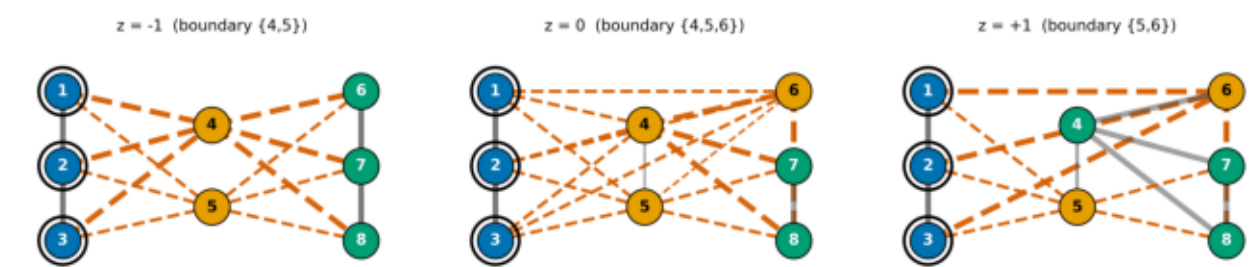


Fig 3: Rate-induced loss of screening

Every frozen configuration along the path has a valid compact ($|B| \leq 3$) exact blanket, yet moving at finite rate transiently and reversibly elevates leakage above 0.

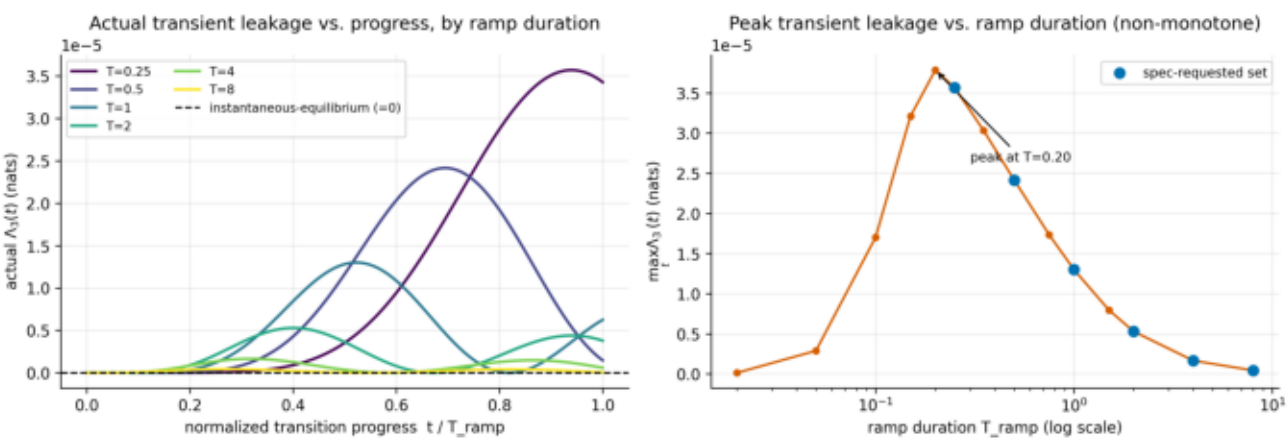


Fig 5: Endpoint validity can hide pathwise failure

Pathwise leakage through control + release, $T=2$ (coordinated control)

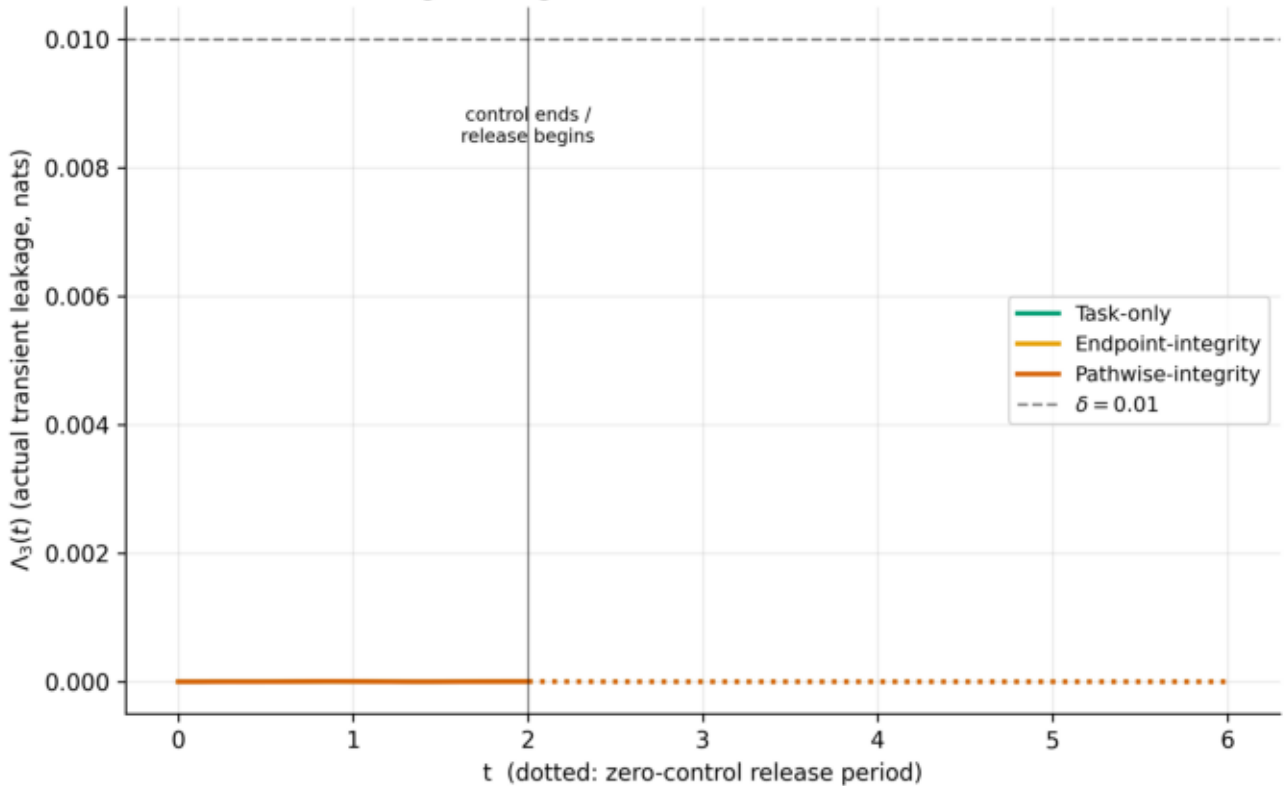


Fig 7: Cost of preserving collective integrity

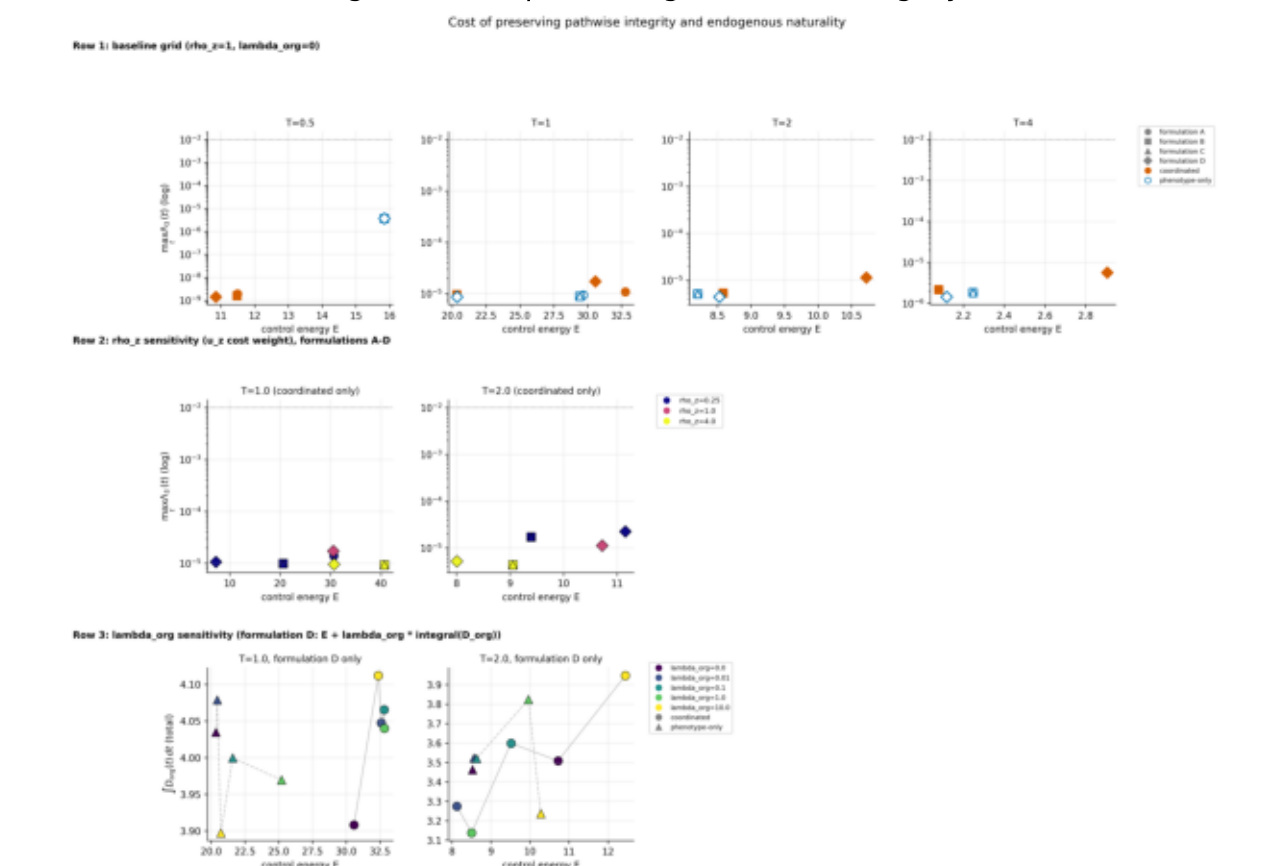


Fig 9: Phenotype-only vs coordinated control

Phenotype-only vs. coordinated control, and sensitivity to the u_z cost weight

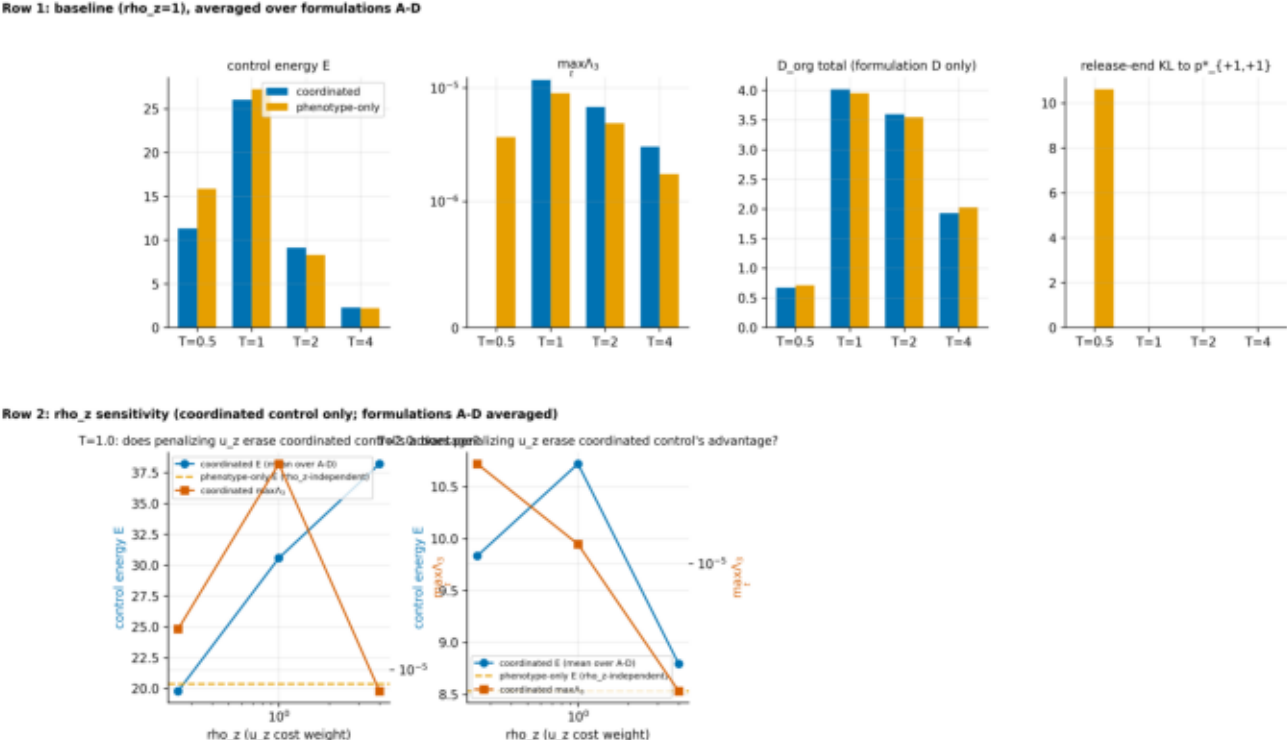


Fig 11: Finite-data integrity classification

Simultaneous-band pathwise classification error vs. sample size ($\delta_{fig11}=1e-05$, 40 realizations/cell, true classes: T_{ramp} in $[0.1, 0.15, 0.2, 0.25]$ vs $[4, 8]$)

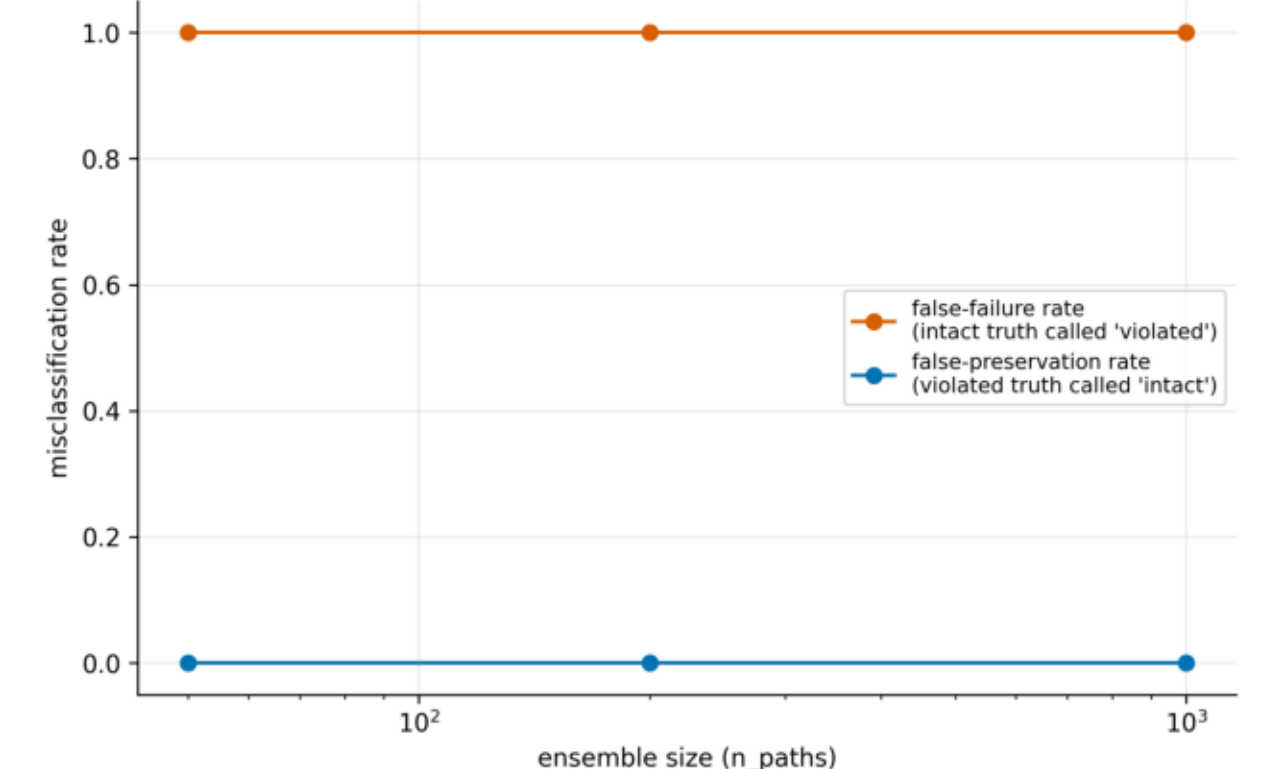


Fig 2: Quasi-static blanket landscape

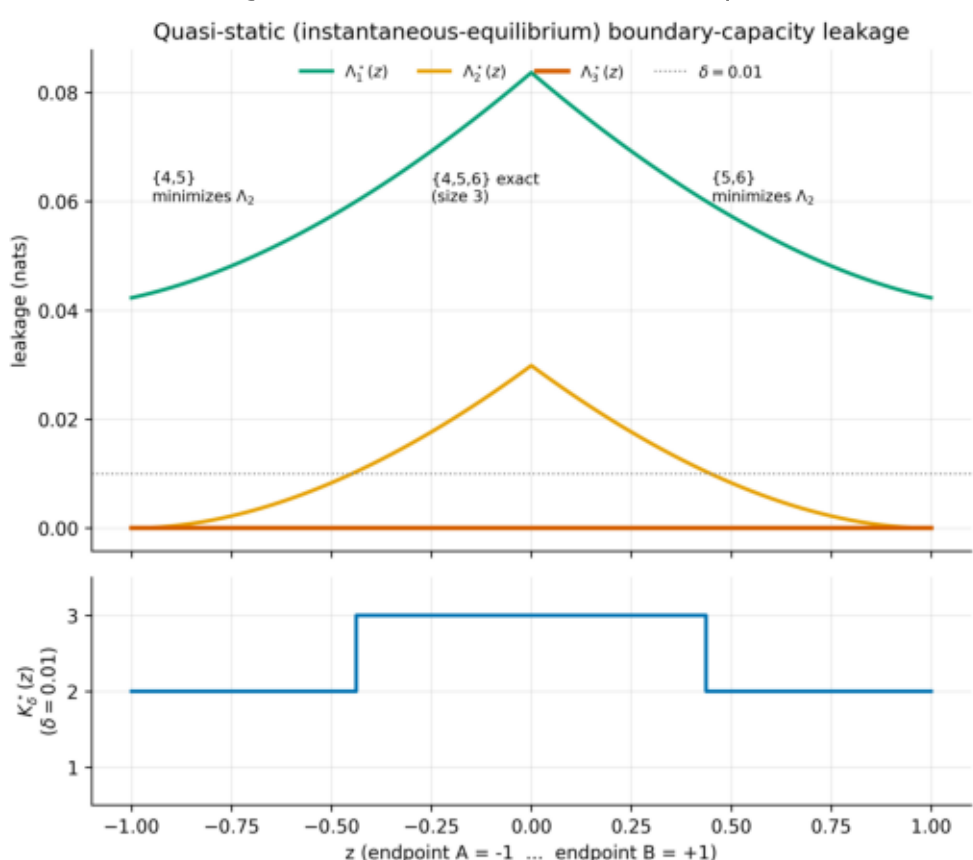


Fig 4: What is being controlled?

Representative protocols at $T=2$ (coordinated control): task-only vs. pathwise-integrity-constrained -- state trajectories and best-found piecewise-linear controls

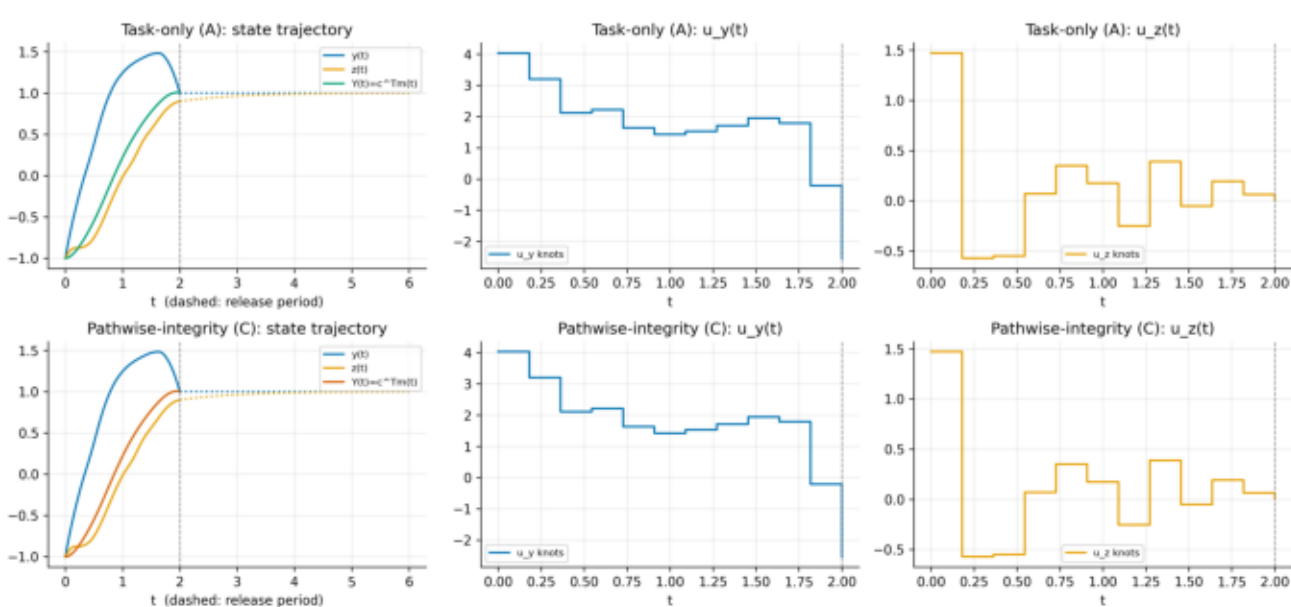


Fig 6 (revised): minimum-cardinality boundary membership

Boundary membership is defined among the smallest boundaries that satisfy the leakage tolerance, so unnecessary zero-leakage supersets are not counted as interface membership.

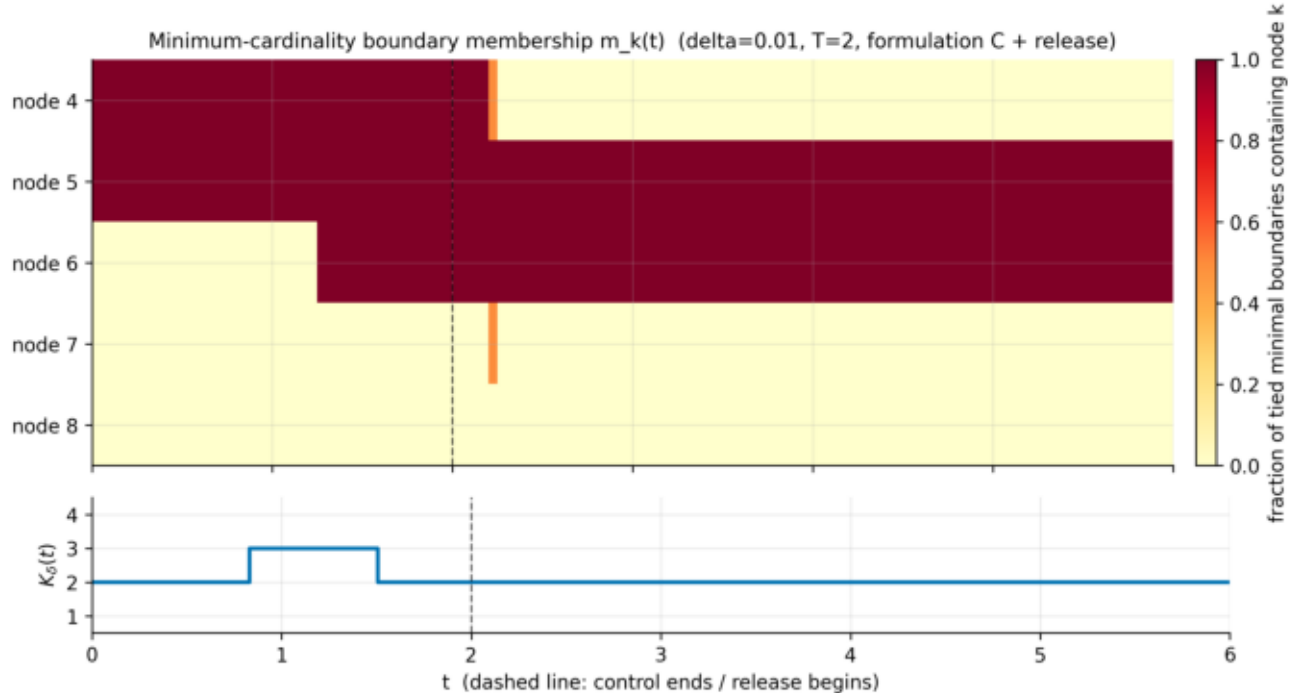


Fig 8 (revised): genuine organization-aware control

Genuine organization-aware control: task-only vs. $\lambda_{org}=1$ vs. $\lambda_{org}=100$ ($T=2$, $\rho_z=1.0$, coordinated)

[TASK NOT MET] flags a protocol whose 'best found' solution did not satisfy the hard task constraints tightly -- its E/D_{org} numbers are not directly comparable to a feasible solution's.

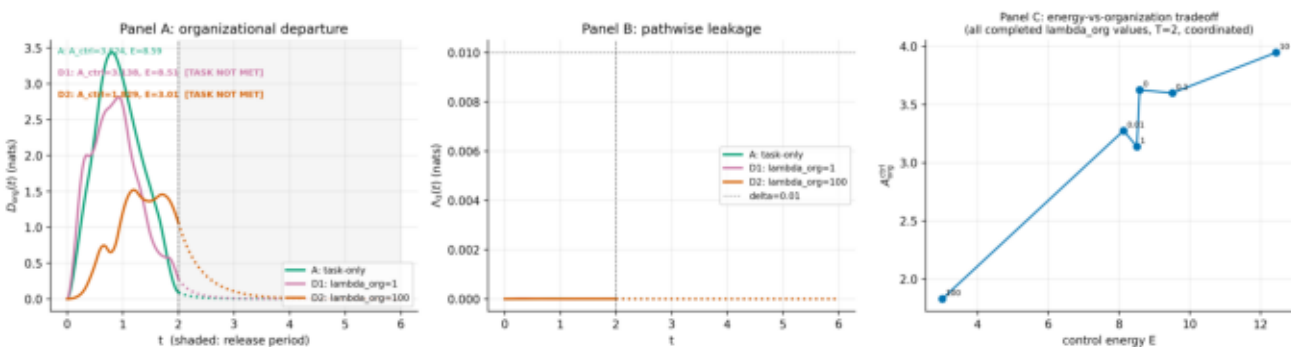


Fig 10: Finite-data recovery

What survives when the oracle covariance is replaced by finite stochastic data (selection/validation split, 100 reps per (n,z))

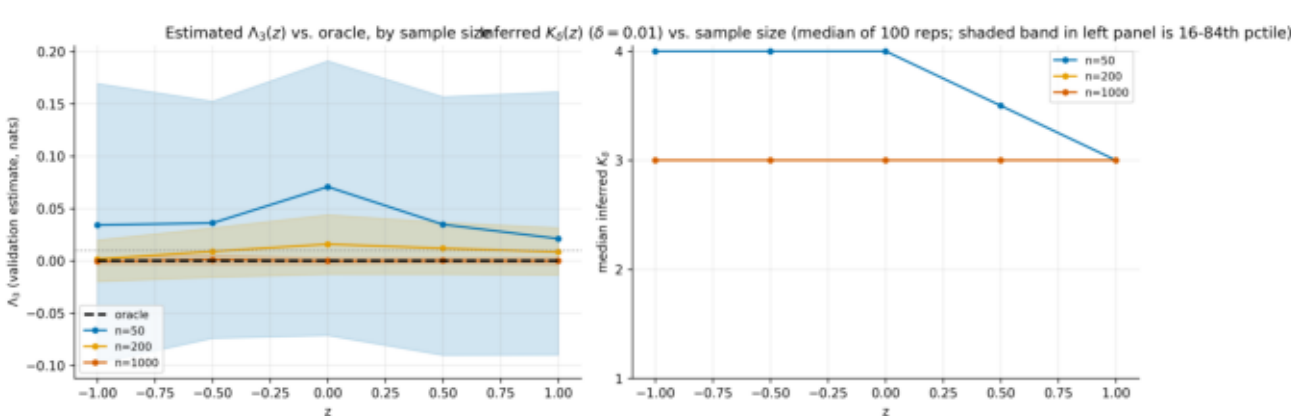


Fig 12: Pathwise constraint activation (unit test)

Constraint-activation unit test using a deliberately stringent tolerance; this is NOT the primary Stage-5 biological/identity threshold.

